

Pawan Kishor Patil

+65 8670 8366
pawanpatil663@gmail.com

Singapore

linkedin.com/in/pawan-kishor-patil
github.com/PawanPatil19

WORK EXPERIENCE

Software Engineer

June 2025 – Present

Venti | Python, Clickhouse, RabbitMQ, Spark, Kubernetes, LangGraph, Cube.js

Singapore

- Architected and led backend development of a company-wide self-serve data platform, building a stateful orchestration engine for multi-step query execution and a translation pipeline compiling natural language into structured queries across heterogeneous internal data sources.
- Led end-to-end migration of core ingestion services from PostgreSQL to ClickHouse, re-architecting the backend for low-latency, horizontally scalable data ingestion across a large vehicle fleet and improving query performance by 10x.
- Built real-time fleet optimization systems: a traffic-aware job scheduling engine improving operational efficiency by 50%, and an ML-based predictive maintenance pipeline for early fault detection across the vehicle fleet.
- Designed a high-throughput storage backend using Apache Iceberg and a distributed Alluxio caching layer in front of object storage, cutting read costs by 40% while keeping service-level latency stable under load.
- Built a client-agnostic deployment architecture with templated Helm charts and agent-driven workflows, cutting new-client onboarding time from days to minutes on shared GCP Kubernetes infrastructure.
- Drove technical direction across backend services, authoring architecture design documents and aligning stakeholders on system boundaries, data flow patterns, and scalability trade-offs.

Full Stack Software Engineer Intern

Jan 2024 – May 2025

Venti | Python, Java, Clickhouse, RESTful APIs, Spark, Git, Pandas, Airflow

Singapore

- Built backend APIs for the simulation platform, enabling engineering teams to trigger vehicle simulations through CI/CD workflows on Airflow, and re-architected response handling with asynchronous workflows to reduce API response times by 70%.
- Re-engineered the vehicle workflow analysis and validation algorithms from the ground up, ensuring 100% metric accuracy and strengthening the correctness of real-time data processing workflows relied on by engineering and operations teams.
- Drove VAPT remediation and security hardening for client-facing applications, identifying and resolving vulnerabilities that enabled the platform to pass an external security audit.
- Built release-readiness dashboards and tools, giving stakeholders a single, real-time view of development and operational metrics to support go or no-go release decisions.

Research Intern

May 2023 – Aug 2023

NUS Smart Systems Institute | C++, Python, PostgreSQL, Swagger, Git

Singapore

- Optimized the YOLOv8 object detection pipeline for a VR application by introducing multithreading and training on custom datasets, improving real-time detection performance and inference responsiveness.
- Built low-latency real-time communication pipelines using WebSockets, WebRTC, and Google Protocol Buffers, reducing server-to-VR headset latency by ~60% and improving responsiveness for interactive VR experiences.
- Built gesture-driven and LLM-assisted interaction workflows within the VR application, integrating LangChain-based pipelines to support more natural user interaction.

Student Assistant

May 2023 – Aug 2023

National University of Singapore | ROS, C++, Flutter

Singapore

- Developed ROS-based control software for a quadruped robot, building application logic to publish movement goals, process velocity commands, and coordinate real-time robot motion through custom control nodes.
- Implemented the communication architecture between the Flutter application, ROS topics, and robotic hardware, enabling low-latency command delivery and telemetry exchange.
- Integrated navigation and safety-control logic, including waypoint handling, joystick inputs, and ultrasonic sensor-based stop mechanisms, to improve reliability during robot operation.

PROJECTS

pg triage tool | Python, FastAPI, MCP

- Built a privacy-preserving Postgres monitoring microservice that materializes stats into periodic SQLite snapshots for LLM context.
- Built a Python MCP Server that dynamically registers Postgres diagnostic tools from a declarative JSON config, so new tools require no code changes. Added per-tool TTL caching to prevent redundant API fan-out during multi-step agent runs.
- Built a GitHub Actions-triggered PR Agent that parses SQL migration files, fetches stats via the MCP Server, and posts structured risk assessments (risk level, row count, build time estimate, pre-migration checks) to GitHub PRs as automated comments with labels.

Distributed Key-Value Store with Raft Consensus | Go

- Implemented the Raft consensus algorithm from scratch, including leader election, log replication, persistence, and snapshotting; verified linearizability under randomized network partitions and node crashes.
- Built a sharded KV store on top of Raft with a separately-replicated shard controller, supporting live shard migration between replica groups without violating linearizable semantics.
- Tested with a chaos harness that injects network partitions and crashes.

EDUCATION

Bachelor of Computing in Computer Science

National University of Singapore

Aug. 2021 – May 2025

Singapore

TECHNICAL SKILLS

Languages: Python, C++, Java, TypeScript, Swift, Go, SQL

Data & Infrastructure: PostgreSQL, ClickHouse, Redis, Kafka, dbt, DuckDB, RabbitMQ, Alluxio, LangGraph, Apache Iceberg, Apache Spark, Airflow, Docker, Kubernetes, Linux

Tools / Frameworks: Next.js, FastAPI, Git, Helm, FastMCP